BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 196 · Arrays, records and tuples

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why must exponents match before adding binary mantissas?

Q2. What does passing an int by value mean?

Q3. What is values[1] for int[] values={4,6,9}?

Q4. a={1,2,3}; a[0]=a[1]+a[2]. Find a.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

array — A fixed-length indexed collection of values of one element type in C#.

index — Zero-based position of an array element.

record — A structure that groups related named fields or properties.

tuple — A compact ordered grouping of multiple values.

assignment — Store a new value at a variable or array location.

New learning

Zero-based indexing — scores[0] is 4 and scores[2] is 2.

Read before writing — The right-hand expression is 4 + 2 = 6.

Replace the selected value — Assign 6 at index 1; the array now contains 4, 6, 2.

Worked example

Trace marks = {5,7,3}; marks[2] = marks[0] + marks[1].

1. Predict — marks[0] + marks[1] means 5 + 7.

2. Confirm and explain — 12 replaces the value at index 2: [5,7,12].

We Do · Practise with guidance

Trace a = {3,9,5}; a[2] = a[0] * 2.

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int[] a = {3,9,5};
a[2] = a[0] * 2;
```

Predict

a[0] is 3, so 3 × 2 = 6.

Confirm and explain

Put 6 into index 2: [3,9,6].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Give the final array after int[] a={8,4,2}; a[1]=a[0]-a[2];

Question 2

When would you choose a record rather than a simple array for a pupil's name and mark?

Question 3

Explain a suitable use of a named tuple in C# and one limitation.

Exit Ticket · Answer before checking

An array with three values has a last index of 3.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The mantissa bit positions must have matching place values.

Q2. The method receives a copy of the integer.

Q3. 6.

Q4. [5,2,3].

You Do pages · Independent questions

Q1. [8,6,2].

Q2. A record groups related named fields with different types, such as string Name and int Mark; a homogeneous array alone does not describe those two fields clearly.

Q3. A tuple such as (string name,int score) can return two related values quickly; a named record is usually clearer when the structure is a long-lived domain entity.

Exit Ticket · Model answer

False: indexes are 0, 1, 2.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: